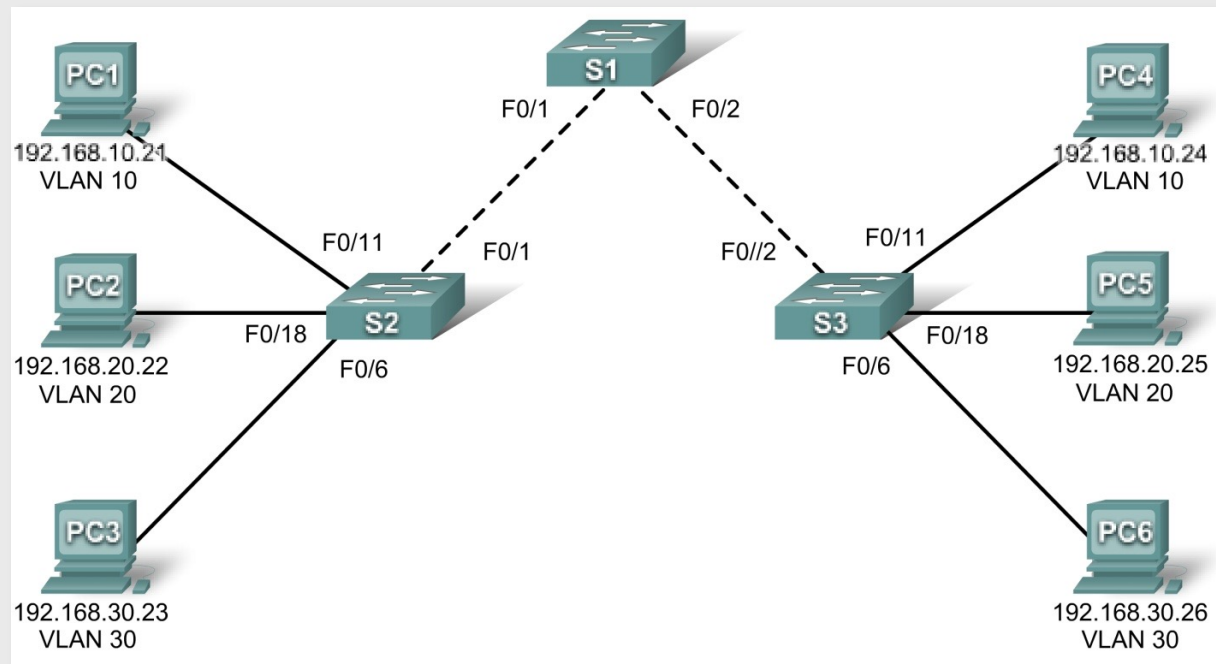


Lab 11

Inter-VLAN networking

Inter-VLAN routing

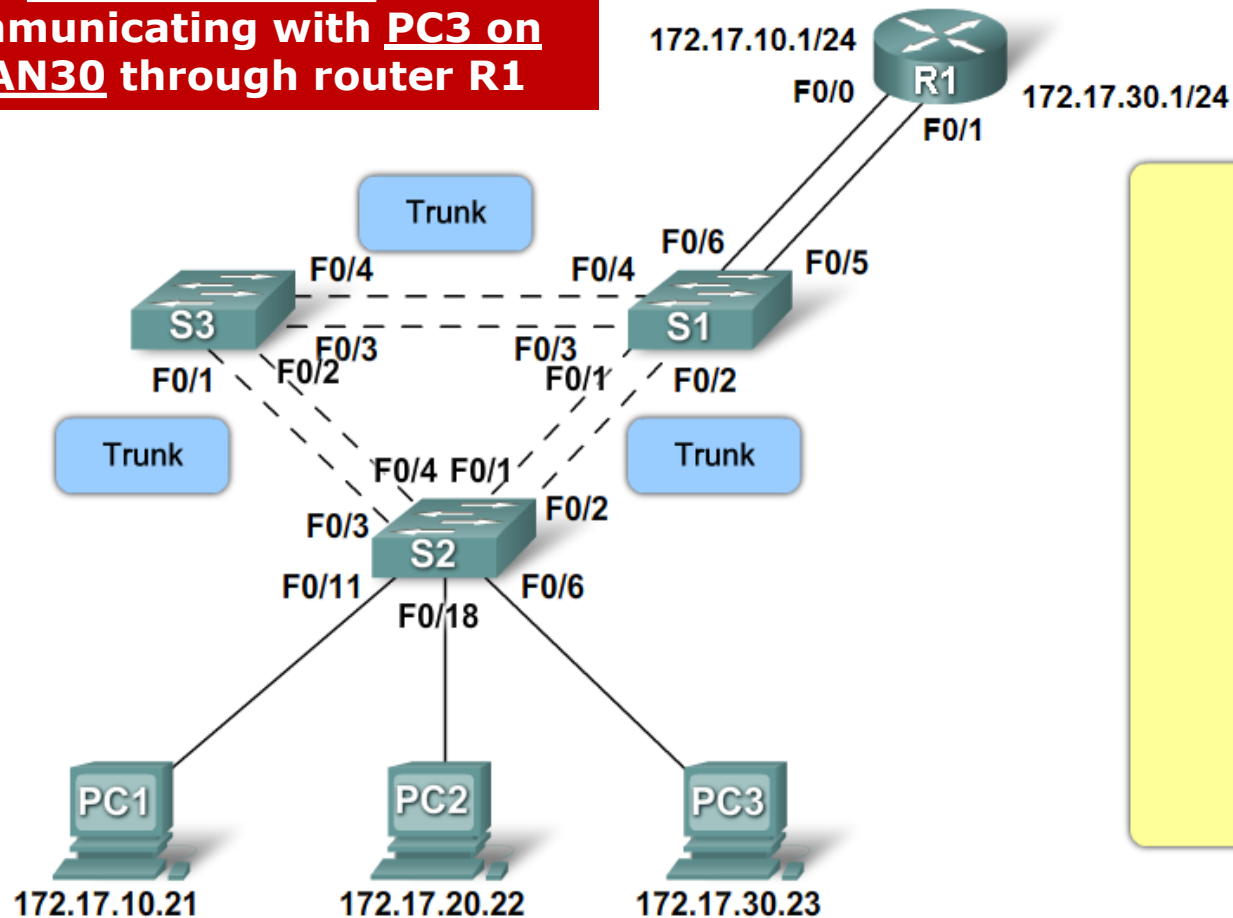
- Each VLAN is a unique broadcast domain
- Hosts on separate VLANs are not able to communicate
- **Inter-VLAN routing:** the process of forwarding network traffic from one VLAN to another VLAN using IP (L3) routing
 - Requires that VLANs are associated with unique IP subnets on the network



Inter-VLAN routing (1)

- Legacy inter-VLAN routing
 - IP router, multiple 'access' links

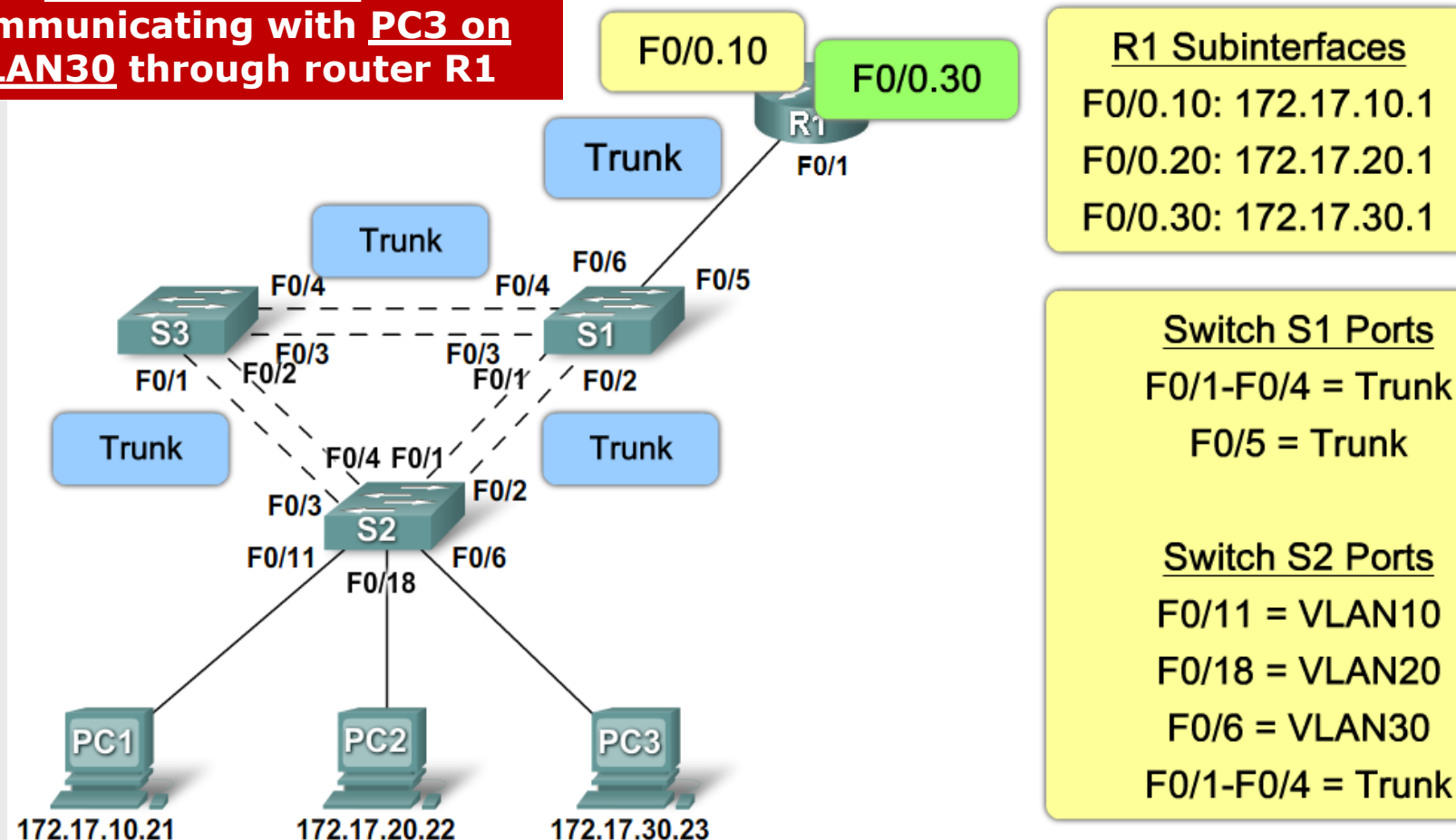
PC1 on VLAN10 is communicating with PC3 on VLAN30 through router R1



Inter-VLAN routing (2)

- "Router-on-a-stick" inter-VLAN routing
 - IP router, single 'trunk' link

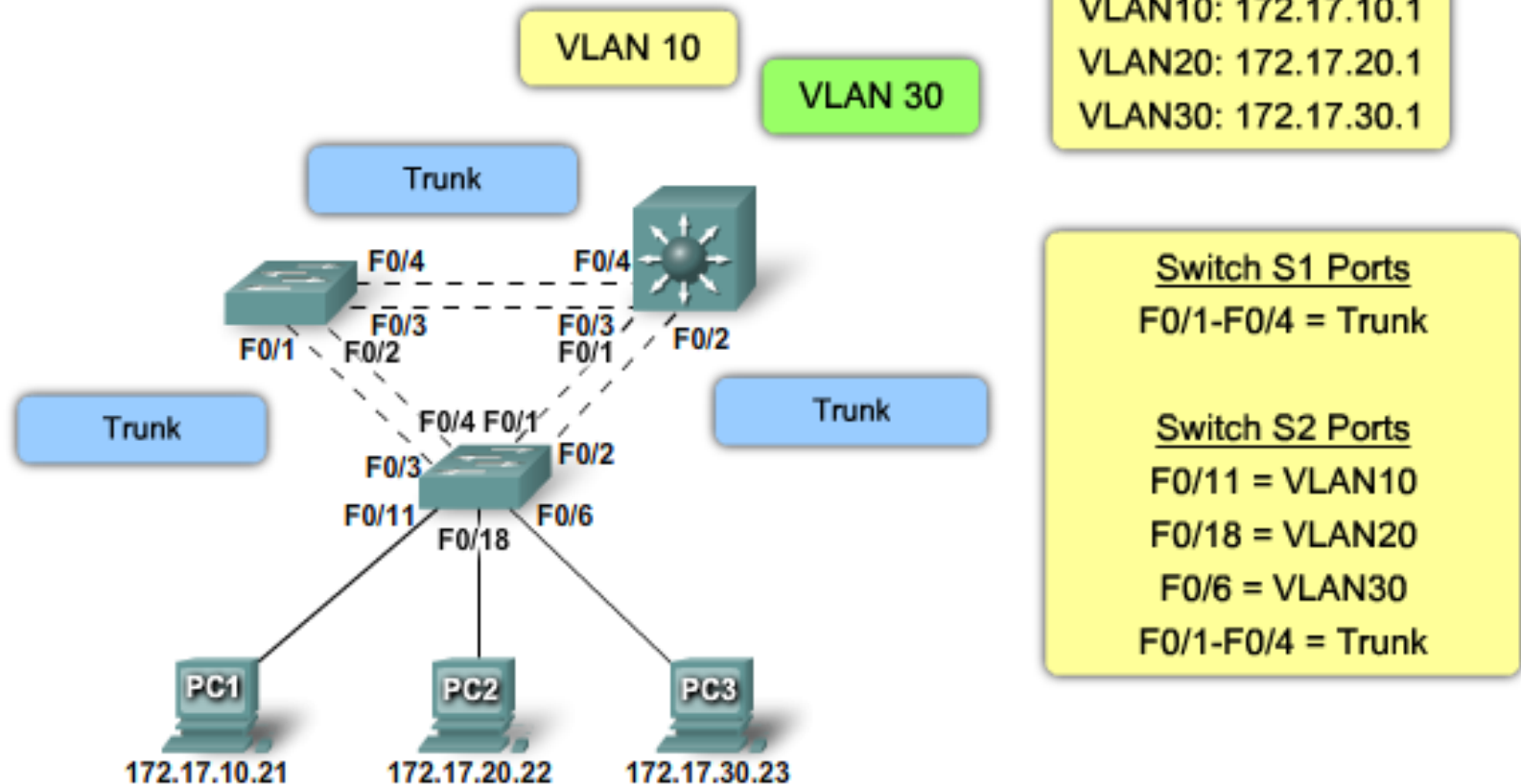
PC1 on VLAN10 is communicating with PC3 on VLAN30 through router R1



Inter-VLAN routing (3)

- Multiple Switch Virtual Interfaces on Multi-Layer Switches

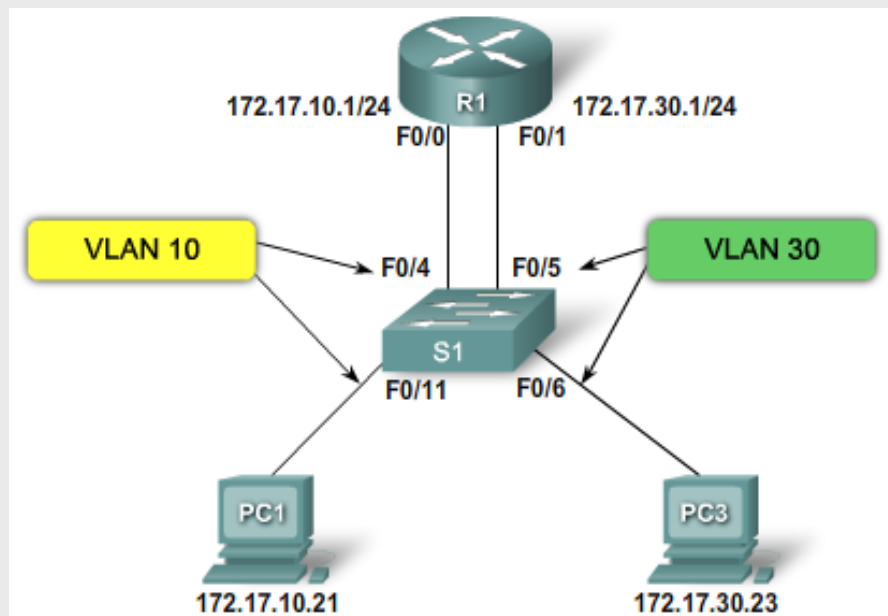
PC1 on VLAN10 is communicating with PC3 on VLAN30 through a MLS



Legacy inter-VLAN routing

```
S1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#vlan 10
S1(config-vlan)#vlan 30
S1(config-vlan)#exit
S1(config)#interface f0/11
S1(config-if)#switchport access vlan 10
S1(config-if)#interface f0/4
S1(config-if)#switchport access vlan 10
S1(config-if)#interface f0/6
S1(config-if)#switchport access vlan 30
S1(config-if)#interface f0/5
S1(config-if)#switchport access vlan 30
S1(config-if)#end
%SYS-5-CONFIG_I: Configured from console by console
```

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface f0/0
R1(config-if)#ip address 172.17.10.1 255.255.255.0
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
R1(config-if)#interface f0/1
R1(config-if)#ip address 172.17.30.1 255.255.255.0
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
R1(config-if)#end
```



```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

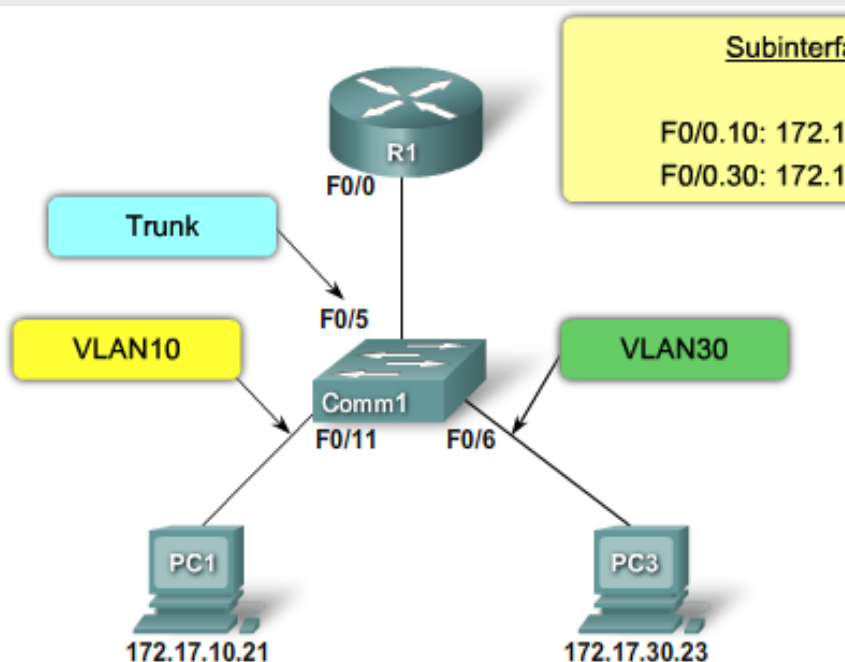
Gateway of last resort is not set

      172.17.0.0/24 is subnetted, 2 subnets
C      172.17.10.0 is directly connected, FastEthernet0/0
C      172.17.30.0 is directly connected, FastEthernet0/1
R1#
```

Router-on-a-stick

```
S1#configure terminal
Enter configuration commands, one per line.
S1(config)#vlan 10
S1(config-vlan)#vlan 30
S1(config-vlan)#exit
S1(config)#interface f0/5
S1(config-if)#switchport mode trunk
S1(config-if)#end
S1#
```

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface f0/0.10
R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip address 172.17.10.1 255.255.255.0
R1(config-subif)#interface f0/0.30
R1(config-subif)#encapsulation dot1q 30
R1(config-subif)#ip address 172.17.30.1 255.255.255.0
R1(config-subif)#interface f0/0
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up
```

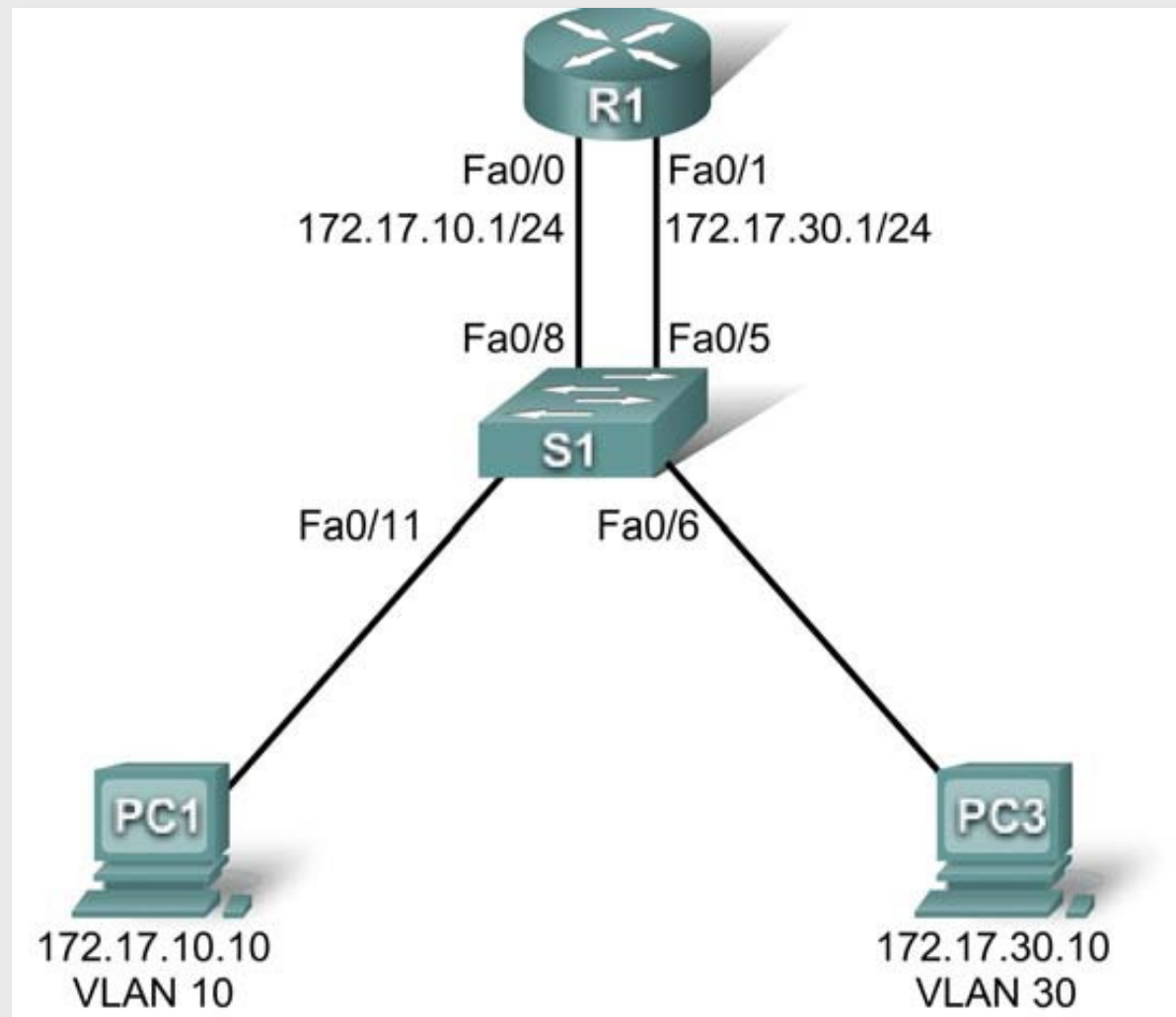


```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

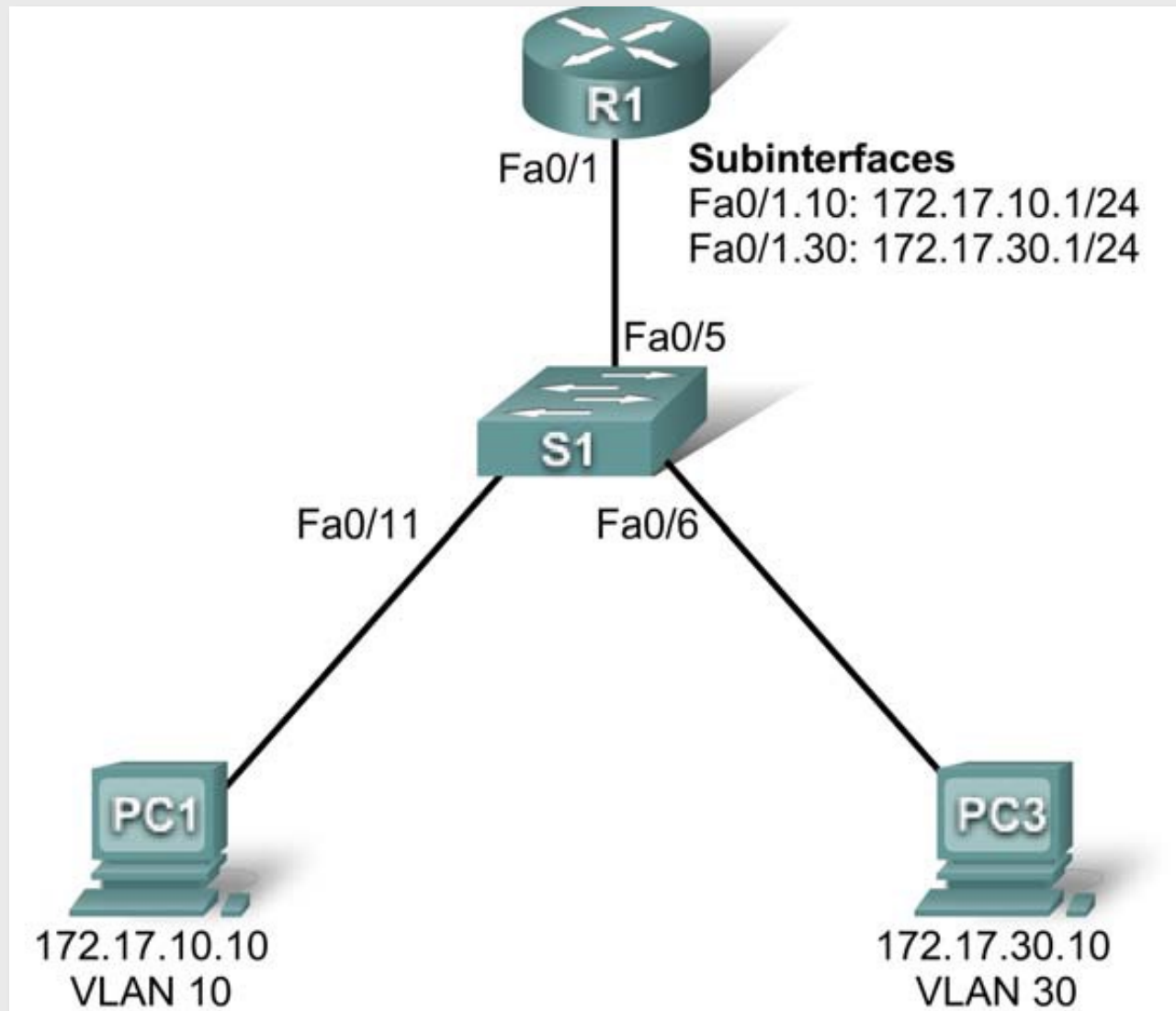
Gateway of last resort is not set

      172.17.0.0/24 is subnetted, 2 subnets
C       172.17.10.0 is directly connected, FastEthernet0/0.10
C       172.17.30.0 is directly connected, FastEthernet0/0.30
R1#
```

Lab activity



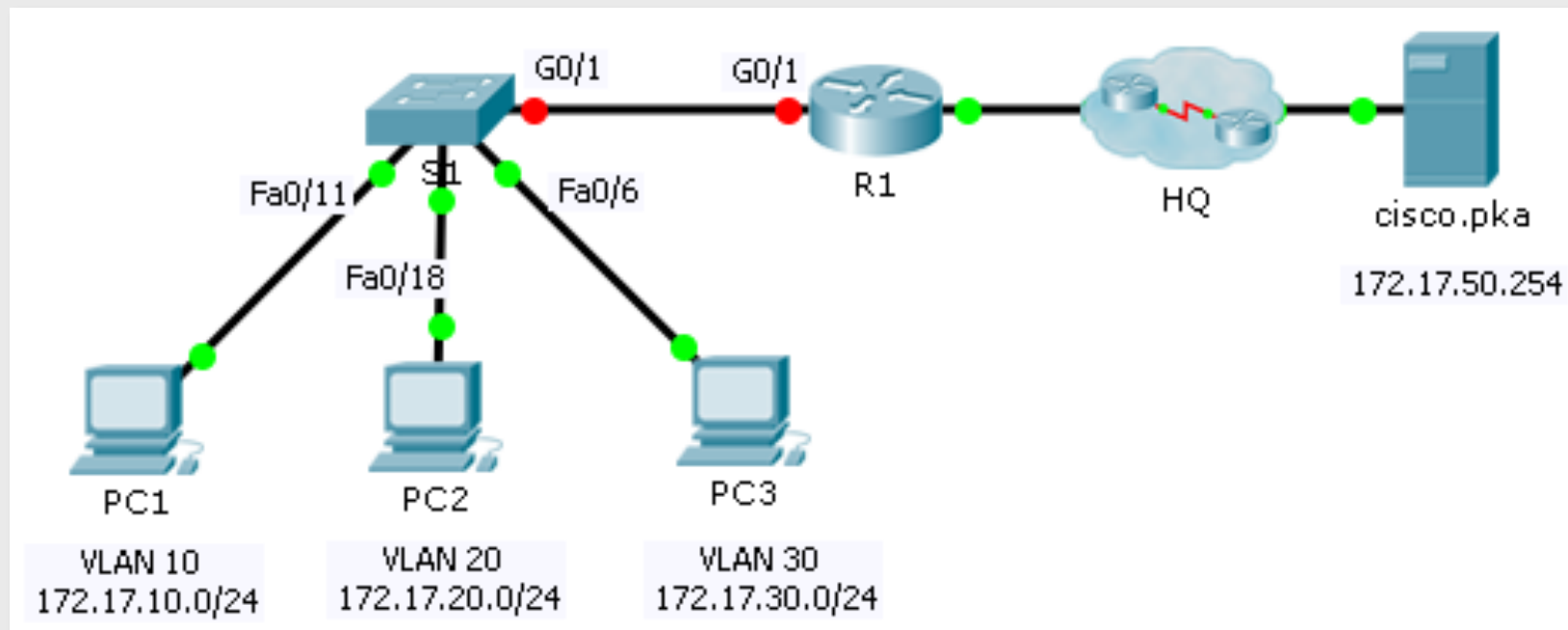
Lab activity



Comparison

Physical Interface	Subinterface
One physical interface per VLAN	One physical interface for many VLANs
No bandwidth contention	Bandwidth contention
Connected to access mode switch port	Connected to trunk mode switch port
More expensive	Less expensive
Less complex connection configuration	More complex connection configuration

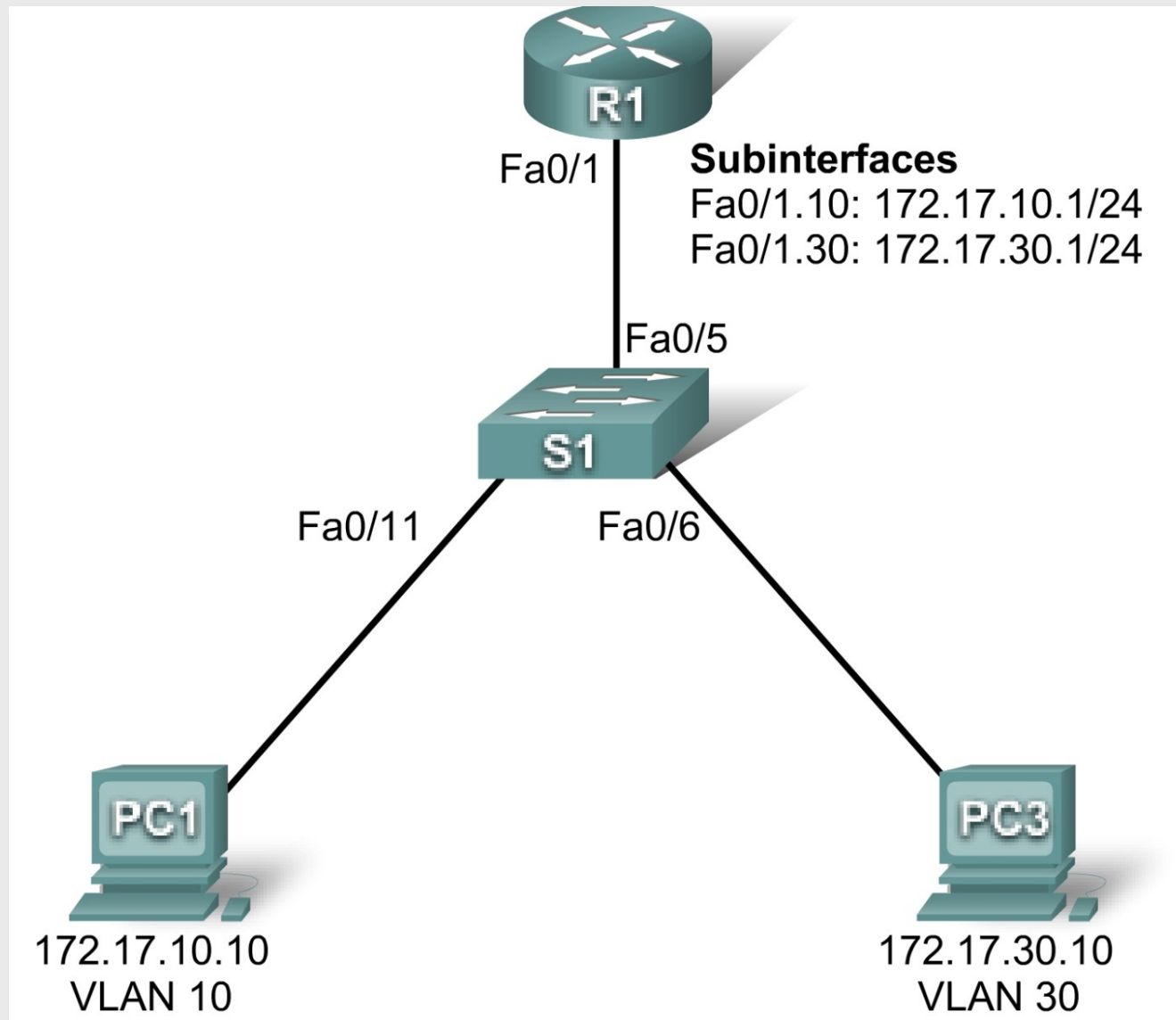
Lab activity



Troubleshooting

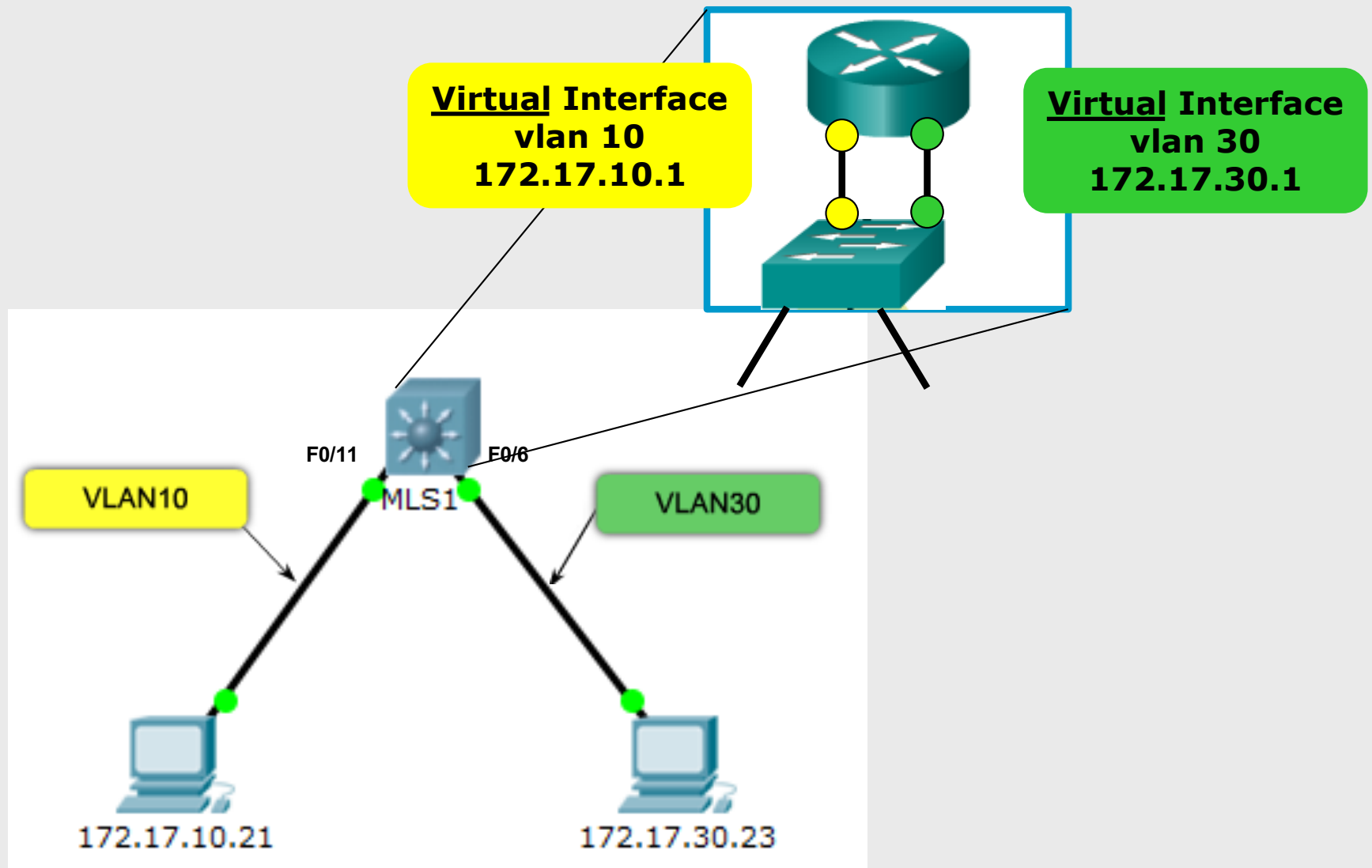
- Switch configuration issues
 - Failing to associate the switch port connected to the router to the right VLAN in traditional inter-VLAN routing
 - Failing to configure the switch port as a trunk in router-on-a-stick inter-VLAN routing
- Router configuration issues
 - Connecting the router physical interface to the wrong switch port
 - Failing to configure 802.1Q encapsulation on the router sub-interface
- IP addressing issues

Lab activity



Layer 3 switching

- Switch Virtual Interface (SVI): provides Layer 3 processing for packets to or from all switch ports associated with that VLAN



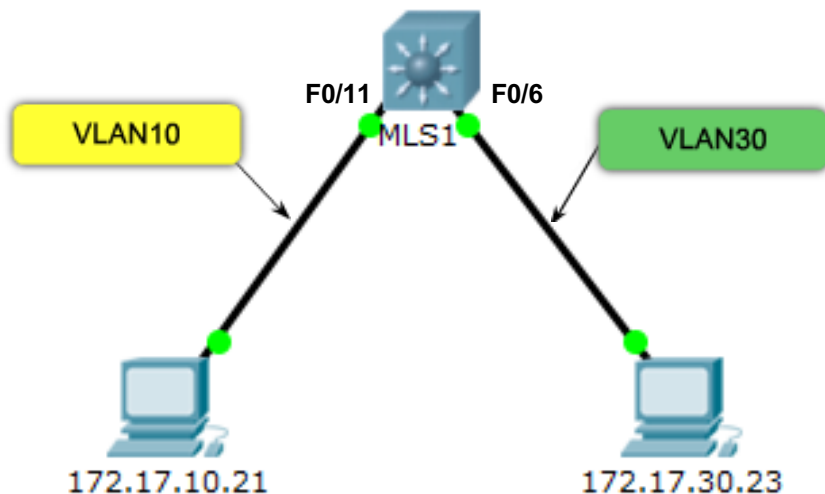
Multi-layer switching

```
MLS1#configure terminal
Enter configuration commands, one per line. En
MLS1(config)#vlan 10
MLS1(config-vlan)#vlan 30
MLS1(config-vlan)#exit
MLS1(config)#interface fa0/11
MLS1(config-if)#switchport mode access
MLS1(config-if)#switchport access vlan 10
MLS1(config-if)#interface fa0/6
MLS1(config-if)#switchport mode access
MLS1(config-if)#switchport access vlan 30
MLS1(config-if)#end
MLS1#
```

```
MLS1#configure terminal
MLS1(config)#ip routing
MLS1(config)#interface vlan 10
MLS1(config)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

MLS1(config-if)#ip address 172.17.10.1 255.255.255.0
MLS1(config-if)#interface Vlan30
MLS1(config)#
%LINK-5-CHANGED: Interface Vlan30, changed state to up

MLS1(config-if)#ip address 172.17.30.1 255.255.255.0
MLS1(config-if)#end
MLS1#
```



```
MLS1#show ip route
```

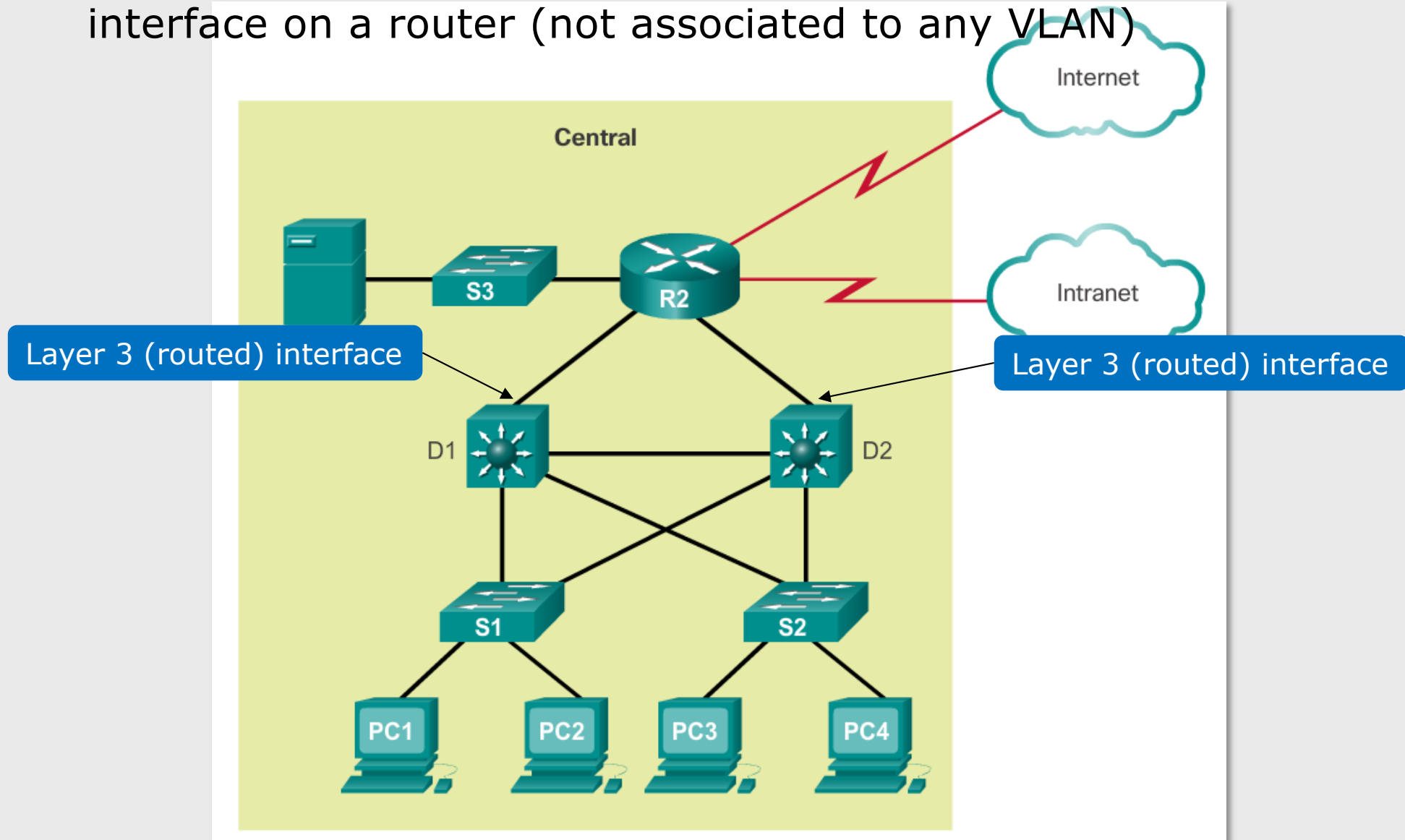
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF internal, N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2, E1 - OSPF external type 1, E2 - OSPF external type 2, i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS level-3, * - candidate default, U - per-user static route, o - OSPF other area, P - periodic downloaded static route

Gateway of last resort is not set

```
172.17.0.0/24 is subnetted, 2 subnets
C      172.17.10.0 is directly connected, Vlan10
C      172.17.30.0 is directly connected, Vlan30
MLS1#
```

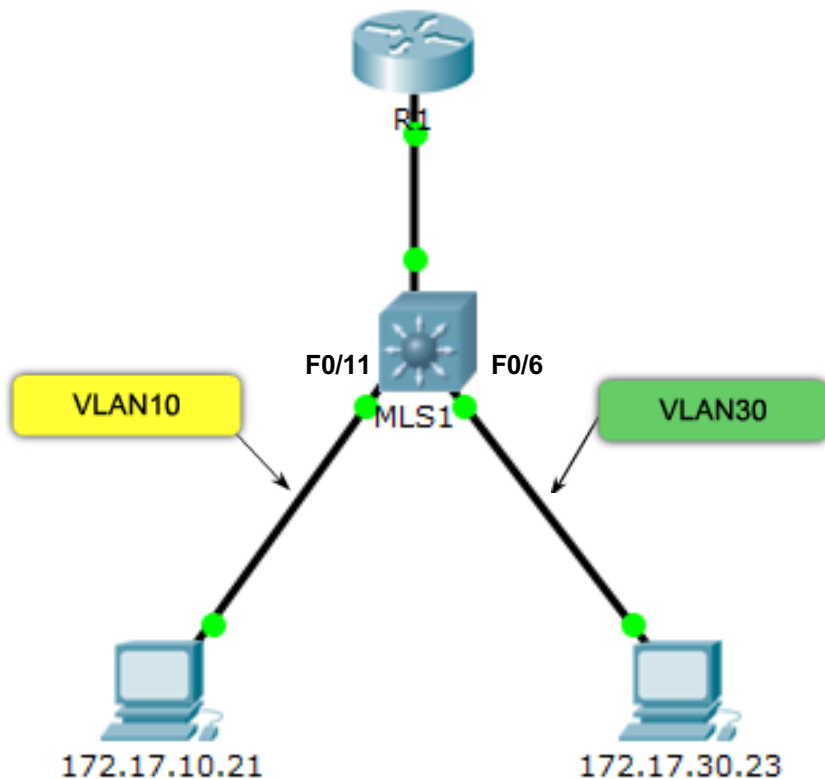
Layer 3 switching

- **Routed port:** A pure Layer 3 interface similar to a physical interface on a router (not associated to any VLAN)



Layer 3 switching

```
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#interface fa0/1
MLS1(config-if)#no switchport
MLS1(config-if)#ip address 192.168.1.2 255.255.255.0
MLS1(config-if)#no shutdown
MLS1(config-if)#end
MLS1#
```



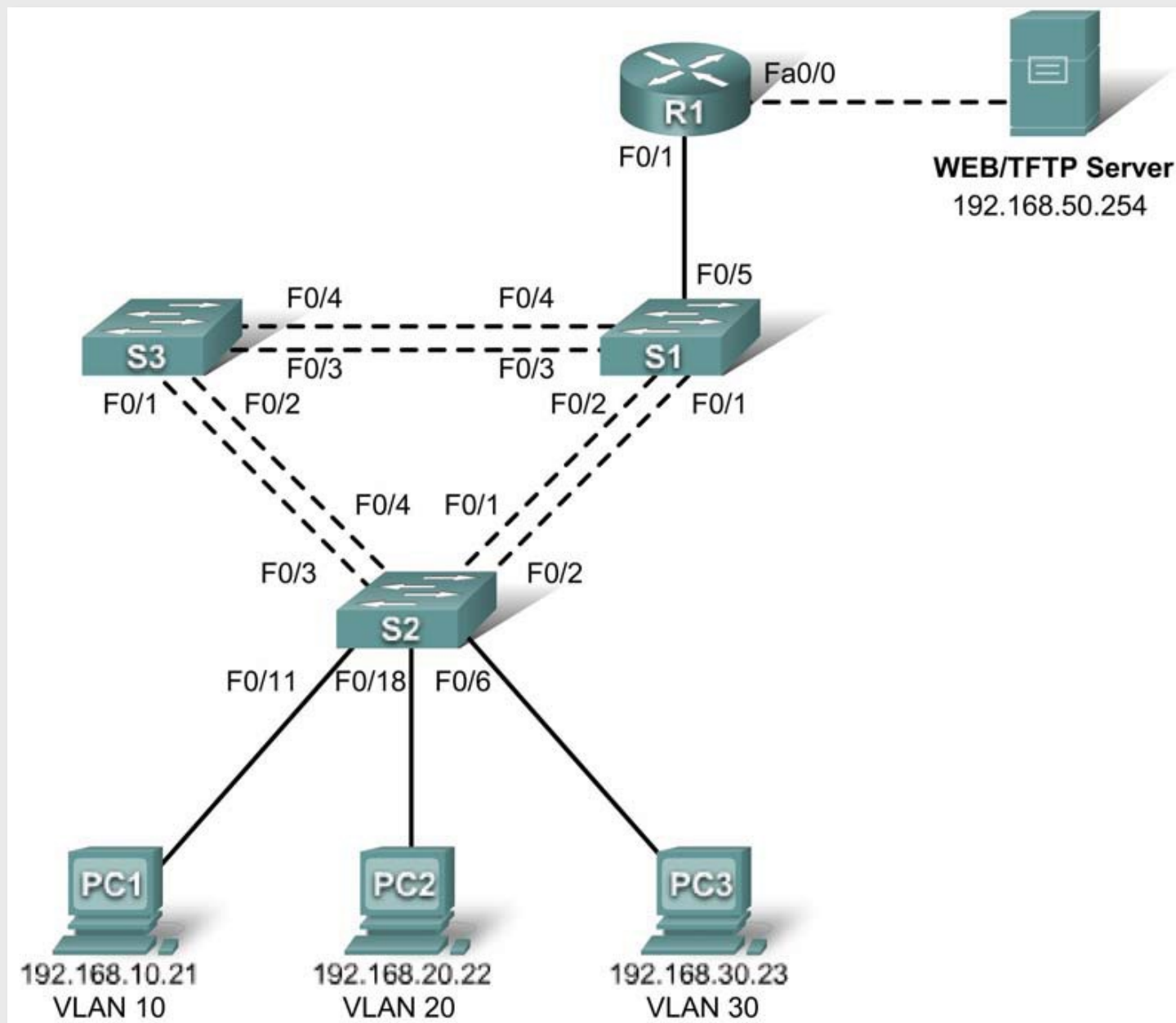
```
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z
MLS1(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.1
MLS1(config-if)#end
MLS1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mob
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF i
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA extern
       E1 - OSPF external type 1, E2 - OSPF external type 2,
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
       * - candidate default, U - per-user static route, o -
       P - periodic downloaded static route
```

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

```
172.17.0.0/24 is subnetted, 2 subnets
C      172.17.10.0 is directly connected, Vlan10
C      172.17.30.0 is directly connected, Vlan30
C      192.168.1.0/24 is directly connected, FastEthernet0/1
S*    0.0.0.0/0 [1/0] via 192.168.1.1
```

```
MLS1#
```

Lab activity



Lab activity

